

**Topic 0: Linear Algebra and Misc***Lecturer: Dr. Kai Liu**Scribe: ChatGPT*

September 1

**Template credit:** UC Berkeley EECS; adapted from the supplied course template.

This lecture develops the linear algebra and calculus behind least squares: matrix products, SVD, positive semidefiniteness, gradients, and projections. It concludes with Taylor expansions and infinite-series examples. All vectors and matrices are real unless explicitly stated otherwise.

**0.1 Matrix multiplication as inner and outer products****0.1.1 Vectors and dimensions**

Vectors are columns. For  $a, b \in \mathbb{R}^d$ , the inner product  $a^\top b$  is a scalar, while the outer product  $ab^\top$  is a  $d \times d$  matrix. More generally,  $ab^\top \in \mathbb{R}^{n \times m}$  for  $a \in \mathbb{R}^n$ ,  $b \in \mathbb{R}^m$ . Every vector is a linear combination of the standard basis vectors; for example,

$$\begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

This idea extends to decomposing a matrix into simple rank-one pieces.

**0.1.2 Two views of the same product**

Let  $A \in \mathbb{R}^{n \times r}$  and  $B \in \mathbb{R}^{m \times r}$ . Then  $C = AB^\top \in \mathbb{R}^{n \times m}$ , and

$$C_{ij} = \sum_{k=1}^r A_{ik} B_{jk} = \langle A_{i,:}, B_{j,:} \rangle. \quad (0.1)$$

Thus an entry is an inner product of two rows. Alternatively, write  $A = [a_1 \cdots a_r]$  and  $B = [b_1 \cdots b_r]$  in terms of their columns. Matrix multiplication also gives

$$AB^\top = \sum_{k=1}^r a_k b_k^\top. \quad (0.2)$$

Each nonzero outer product has rank one. The two formulas describe the same operation at the entry level and at the matrix level, respectively.

**Worked example.** With  $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$  and  $B = \begin{pmatrix} 3 & 2 \\ 1 & 2 \end{pmatrix}$ ,

$$AB^\top = \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \end{pmatrix} \begin{pmatrix} 2 & 2 \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ 9 & 3 \end{pmatrix} + \begin{pmatrix} 4 & 4 \\ 8 & 8 \end{pmatrix} = \begin{pmatrix} 7 & 5 \\ 17 & 11 \end{pmatrix}.$$

For instance, the lower-left entry is  $3 \cdot 3 + 4 \cdot 2 = 17$ . Keeping track of the transpose prevents confusing  $AB^\top$  with  $AB$ .

## 0.2 Norms, trace, and quadratic expansions

### 0.2.1 Vector and matrix norms

For  $v \in \mathbb{R}^d$  and  $1 \leq p < \infty$ ,

$$\|v\|_p = \left( \sum_{i=1}^d |v_i|^p \right)^{1/p}, \quad \|v\|_\infty = \max_i |v_i|.$$

In particular,  $\|v\|_2^2 = v^\top v$ . The trace of a square matrix is the sum of its diagonal entries, so

$$\|v\|_2^2 = v^\top v = \text{tr}(v^\top v) = \text{tr}(vv^\top). \quad (0.3)$$

Here  $v^\top v$  is a scalar (a  $1 \times 1$  matrix), whereas  $vv^\top$  is a  $d \times d$  matrix. For  $v = (2, 6)^\top$ ,

$$vv^\top = \begin{pmatrix} 4 & 12 \\ 12 & 36 \end{pmatrix}, \quad \|v\|_2^2 = 4 + 36 = 40.$$

For matrices  $M, N \in \mathbb{R}^{n \times m}$ , define the Frobenius inner product and norm:

$$\langle M, N \rangle_F = \sum_{i,j} M_{ij} N_{ij} = \text{tr}(M^\top N), \quad \|M\|_F^2 = \text{tr}(M^\top M). \quad (0.4)$$

The subscript  $F$  distinguishes this matrix norm from the vector Euclidean norm.

### 0.2.2 Cyclic trace identity does not mean commutativity

If  $A \in \mathbb{R}^{n \times m}$  and  $B \in \mathbb{R}^{m \times n}$ , then

$$\text{tr}(AB) = \sum_{i=1}^n \sum_{j=1}^m A_{ij} B_{ji} = \text{tr}(BA). \quad (0.5)$$

The two products may have different sizes, and even when their sizes agree, the matrices themselves generally differ. For example,

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix}, \quad AB = \begin{pmatrix} 6 & 8 \\ 10 & 13 \end{pmatrix}, \quad BA = \begin{pmatrix} 5 & 8 \\ 9 & 14 \end{pmatrix}.$$

Both traces equal 19. For compatible products, cyclic rotation also gives  $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$ ; it does not permit arbitrary reordering.

### 0.2.3 Expanding a squared norm

The familiar scalar identity  $(a + b)^2 = a^2 + 2ab + b^2$  becomes

$$\boxed{\|a + b\|_2^2 = \|a\|_2^2 + 2a^\top b + \|b\|_2^2}. \quad (0.6)$$

Indeed, expand  $(a + b)^\top (a + b)$  and use  $b^\top a = a^\top b$ . For  $a = (1, 1)^\top$ ,  $b = (2, 2)^\top$ , the two sides are 18 and  $2 + 8 + 8$ . If  $a^\top b = 0$ , the cross term vanishes, giving the Pythagorean identity. The same expansion holds for matrices with the Frobenius inner product.

### 0.3 Symmetric positive semidefinite matrices

A square matrix  $Q \in \mathbb{R}^{d \times d}$  is symmetric if  $Q^\top = Q$ . The following notions concern real *symmetric* matrices.

**Definition 0.1 (PSD and PD)** A symmetric matrix  $Q$  is *positive semidefinite (PSD)*, written  $Q \succeq 0$ , if  $v^\top Q v \geq 0$  for every  $v \in \mathbb{R}^d$ . It is *positive definite (PD)*, written  $Q \succ 0$ , if  $v^\top Q v > 0$  for every nonzero  $v$ .

The abbreviation SPSD makes the symmetry requirement explicit. The difference between  $\geq 0$  and  $> 0$  determines whether nonzero zero-curvature directions are allowed. Also, testing one vector is not enough to prove PSD; a single negative value is enough to disprove it.

#### 0.3.1 Symmetry alone is insufficient

The matrix

$$Q_1 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

is symmetric and has positive entries, but for  $v = (-2, 1)^\top$ ,  $Q_1 v = (0, -3)^\top$  and  $v^\top Q_1 v = -3 < 0$ . It is not PSD. In contrast,

$$Q_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

has eigenvalues 3 and 1 and is PD. For the same  $v$ , its quadratic form equals 6, but positive definiteness follows from all eigenvalues being positive, not from that one test.

#### 0.3.2 The eigenvalue characterization

By the spectral theorem, a real symmetric matrix has an eigendecomposition  $Q = U\Lambda U^\top$ , where  $U^\top U = I$  and  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ . Consequently,

$$v^\top Q v = \sum_{i=1}^d \lambda_i (u_i^\top v)^2. \quad (0.7)$$

If all  $\lambda_i \geq 0$ , the sum is nonnegative. Conversely, setting  $v = u_i$  gives  $v^\top Q v = \lambda_i$ , so PSD implies all  $\lambda_i \geq 0$ . Strict positivity of every eigenvalue similarly characterizes PD.

#### 0.3.3 Gram matrices are always PSD

For any  $A \in \mathbb{R}^{n \times d}$ , the matrix  $Q = A^\top A$  is symmetric and

$$v^\top Q v = v^\top A^\top A v = (Av)^\top (Av) = \|Av\|_2^2 \geq 0. \quad (0.8)$$

It is PD exactly when  $Av = 0$  implies  $v = 0$ , that is, when  $A$  has full column rank. For example, if  $Av = (2, 6)^\top$ , this quadratic form equals 40.

**Weighted squared lengths.** For symmetric PSD  $Q$ , write  $\|x\|_Q^2 = x^\top Q x$ . Then

$$\|a + b\|_Q^2 = \|a\|_Q^2 + 2a^\top Q b + \|b\|_Q^2.$$

The square root is a norm when  $Q \succ 0$ . For singular PSD  $Q$ , it is a seminorm: a nonzero vector in  $\text{null}(Q)$  has zero length.

## 0.4 The singular value decomposition

Let  $X \in \mathbb{R}^{n \times m}$ ,  $q = \min\{n, m\}$ , and  $r = \text{rank}(X)$ . The SVD separates the action of  $X$  into orthogonal coordinate changes and nonnegative scaling. Unlike an eigendecomposition, it applies to rectangular matrices and does not require symmetry.

### 0.4.1 Full, economy-size, and compact forms

The full SVD is

$$X = U \Sigma V^\top, \quad U \in \mathbb{R}^{n \times n}, \quad \Sigma \in \mathbb{R}^{n \times m}, \quad V \in \mathbb{R}^{m \times m}. \quad (0.9)$$

The matrices  $U$  and  $V$  are orthogonal, and the rectangular diagonal matrix  $\Sigma$  has diagonal entries  $\sigma_1 \geq \dots \geq \sigma_q \geq 0$ . Exactly  $r$  singular values are positive.

Keeping only the positive singular values gives the *compact SVD*:

$$\boxed{X = U_r \Sigma_r V_r^\top}, \quad U_r \in \mathbb{R}^{n \times r}, \quad \Sigma_r \in \mathbb{R}^{r \times r}, \quad V_r \in \mathbb{R}^{m \times r}. \quad (0.10)$$

Here  $\Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r)$  and  $U_r^\top U_r = V_r^\top V_r = I_r$ . In general,  $U_r U_r^\top \neq I_n$ ; it is the projector onto  $\text{col}(X)$ .

| SVD convention    | Left factor  | Middle factor | Right factor before transpose |
|-------------------|--------------|---------------|-------------------------------|
| Full              | $n \times n$ | $n \times m$  | $m \times m$                  |
| Economy-size      | $n \times q$ | $q \times q$  | $m \times q$                  |
| Compact, rank $r$ | $n \times r$ | $r \times r$  | $m \times r$                  |

The notes call the rank- $r$  form “reduced SVD.” Software and textbooks sometimes use “reduced” for the economy-size form, which keeps  $q$  directions including any zero singular values. The shapes, rather than the name alone, resolve this ambiguity. If  $X = 0$ , its compact expansion is the empty sum.

### 0.4.2 SVD as a sum of rank-one matrices

Using (0.2),

$$X = [u_1 \ \dots \ u_r] \Sigma_r [v_1 \ \dots \ v_r]^\top = \sum_{i=1}^r \sigma_i u_i v_i^\top. \quad (0.11)$$

Each term has size  $n \times m$ . For an input  $z \in \mathbb{R}^m$ ,

$$Xz = \sum_{i=1}^r \sigma_i u_i (v_i^\top z).$$

First take the coordinates  $v_i^\top z$ , then scale them by  $\sigma_i$ , and finally assemble the output along  $u_i$ . In particular,

$$Xv_i = \sigma_i u_i, \quad X^\top u_i = \sigma_i v_i. \quad (0.12)$$

These identities also imply  $X^\top X v_i = \sigma_i^2 v_i$  and  $XX^\top u_i = \sigma_i^2 u_i$ .

### 0.4.3 Orthogonality of the rank-one components

The singular vectors are orthonormal, but so are the rank-one matrices  $u_i v_i^\top$  under the Frobenius inner product. Indeed,

$$\begin{aligned}\langle u_i v_i^\top, u_j v_j^\top \rangle_F &= \text{tr}((u_i v_i^\top)^\top u_j v_j^\top) \\ &= \text{tr}(v_i u_i^\top u_j v_j^\top) = (u_i^\top u_j)(v_j^\top v_i) \\ &= \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}\end{aligned}\tag{0.13}$$

Thus each component has Frobenius norm one. Expanding the squared norm of the SVD sum and canceling the cross terms gives

$$\|X\|_F^2 = \sum_{i=1}^r \sigma_i^2.\tag{0.14}$$

If  $X_k = \sum_{i=1}^k \sigma_i u_i v_i^\top$  for  $k < r$ , the same calculation gives  $\|X - X_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2$ . This identity quantifies the error of truncating the SVD; no separate optimization theorem is needed to establish the error of this particular approximation.

### 0.4.4 Why singular vectors are not unique

Replacing both  $u_i$  and  $v_i$  by their negatives leaves a component unchanged:

$$\sigma_i(-u_i)(-v_i)^\top = \sigma_i u_i v_i^\top.$$

Therefore two valid SVD implementations may return opposite signs for the same singular-vector pair. Flipping only one vector would change the component.

Repeated singular values permit a larger ambiguity. Within a block of equal singular value  $\sigma$ , let the corresponding factors be  $U_b, V_b$ . For any orthogonal matrix  $R$  of the block size,

$$(U_b R)(\sigma I)(V_b R)^\top = \sigma U_b R R^\top V_b^\top = \sigma U_b V_b^\top.$$

The subspace is determined, but a particular orthonormal basis within it is not. Singular vectors attached to zero singular values have further freedom.

### 0.4.5 SVD and symmetric eigendecomposition

If  $Q \succeq 0$  is symmetric, its eigendecomposition  $Q = U \Lambda U^\top$  has nonnegative eigenvalues and is also an SVD. Thus one can choose the left and right singular vectors to be the same. For a symmetric *indefinite* matrix, eigenvalues may be negative but singular values remain nonnegative: they are the absolute values of the eigenvalues. In that case  $Q = U \Sigma U^\top$  with  $\Sigma \geq 0$  would incorrectly imply PSD.

**A small example.**  $Q = \text{diag}(2, -1)$  has eigenvalues 2,  $-1$  but singular values 2, 1. One SVD is  $U = \text{diag}(1, -1)$ ,  $\Sigma = \text{diag}(2, 1)$ , and  $V = I_2$ .

## 0.5 Linear regression in matrix form

### 0.5.1 From a fitted line to a design matrix

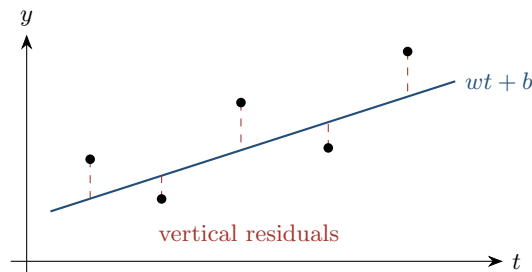
Given pairs  $(t_i, y_i)$ ,  $i = 1, \dots, n$ , fit the affine model  $\hat{y}_i = wt_i + b$ . Its total squared error is

$$\sum_{i=1}^n (y_i - (wt_i + b))^2.$$

Introduce

$$A = \begin{pmatrix} t_1 & 1 \\ t_2 & 1 \\ \vdots & \vdots \\ t_n & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} w \\ b \end{pmatrix}, \quad y = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}. \quad (0.15)$$

Then  $A\beta$  contains all fitted responses, and the sum becomes  $\|A\beta - y\|_2^2$ . The last column of ones represents the intercept. For multiple features, use any design matrix  $A \in \mathbb{R}^{n \times d}$  and a coefficient vector  $\beta \in \mathbb{R}^d$ .



The diagram is schematic. Ordinary least squares measures vertical response errors when predictors are held fixed; it does not minimize perpendicular distances from points to the line in the  $(t, y)$  plot.

### 0.5.2 The optimization objective

We use the convenient normalization

$$\min_{\beta \in \mathbb{R}^d} f(\beta), \quad f(\beta) = \frac{1}{2} \|A\beta - y\|_2^2. \quad (0.16)$$

Multiplying an objective by a positive constant does not change its minimizers. The factor  $1/2$  simply cancels the factor of two when differentiating. Without it, the gradient is twice as large, which matters for a fixed gradient-descent step size even though the minimizers agree.

The handwritten notes use both  $X$  and  $A$  for the design matrix. We use  $A$  from this point onward and reserve  $\beta$  for the coefficients, so input coordinates are not confused with the optimization variable.

## 0.6 Gradients from small perturbations

### 0.6.1 The scalar idea

For  $g(x) = x^2$ , direct expansion gives

$$g(x+h) = (x+h)^2 = x^2 + 2xh + h^2.$$

Subtract  $g(x)$ , divide by  $h \neq 0$ , and let  $h \rightarrow 0$ :

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} (2x + h) = 2x.$$

The derivative is the coefficient of the first-order change. The remainder  $h^2$  is negligible compared with  $|h|$  as  $h \rightarrow 0$ .

For the lecture's scalar minimization example,

$$q(x) = \frac{1}{2}x^2 - 3x + 1, \quad q'(x) = x - 3, \quad q''(x) = 1.$$

The stationary point is  $x = 3$ , and completing the square verifies that it is the unique global minimizer:

$$q(x) = \frac{1}{2}(x-3)^2 - \frac{7}{2}.$$

In general, a zero derivative is a necessary condition for an unconstrained interior minimum of a differentiable function, not automatically a sufficient one. For example,  $-x^2$  also has zero derivative at zero, but zero is a maximum.

### 0.6.2 The vector definition

A function  $f: \mathbb{R}^d \rightarrow \mathbb{R}$  is differentiable at  $x$  if there is a vector  $\nabla f(x) \in \mathbb{R}^d$  such that

$$f(x+h) = f(x) + \langle \nabla f(x), h \rangle + o(\|h\|_2). \quad (0.17)$$

The notation  $o(\|h\|_2)$  means that the remainder divided by  $\|h\|_2$  tends to zero. The gradient collects the partial derivatives:

$$\nabla f(x) = \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_d \end{pmatrix}.$$

For a direction  $v$ , differentiability implies  $\frac{d}{dt} f(x+tv)|_{t=0} = \nabla f(x)^\top v$ .

### 0.6.3 Two basic identities

If  $f(x) = a^\top x$ , then  $f(x+h) - f(x) = a^\top h$  exactly, so  $\nabla f(x) = a$ . If  $f(x) = \frac{1}{2} \|x\|_2^2$ , then

$$f(x+h) = f(x) + x^\top h + \frac{1}{2} \|h\|_2^2, \quad \nabla f(x) = x.$$

These are the vector versions of differentiating a linear function and a quadratic. They also show why the gradient has the same dimension as  $x$ .

## 0.7 Deriving the least-squares gradient and Hessian

Let  $f(\beta) = \frac{1}{2} \|A\beta - y\|_2^2$  and perturb  $\beta$  by  $h \in \mathbb{R}^d$ . Using (0.6),

$$\begin{aligned} f(\beta + h) &= \frac{1}{2} \|A\beta - y + Ah\|_2^2 \\ &= \frac{1}{2} \|A\beta - y\|_2^2 + (A\beta - y)^\top Ah + \frac{1}{2} \|Ah\|_2^2 \\ &= f(\beta) + \langle A^\top (A\beta - y), h \rangle + \frac{1}{2} h^\top A^\top Ah. \end{aligned} \quad (0.18)$$

The last term is second order in  $h$ . Comparing the linear term with the gradient definition gives

$$\boxed{\nabla f(\beta) = A^\top (A\beta - y)}, \quad \boxed{\nabla^2 f(\beta) = A^\top A}. \quad (0.19)$$

The expansion is exact because  $f$  is quadratic; there are no terms of order three or higher. Also,  $A^\top (A\beta - y)$  has size  $d \times 1$ , as required.

### 0.7.1 The same result from the chain rule

Set  $r(\beta) = A\beta - y$  and  $g(r) = \frac{1}{2} \|r\|_2^2$ . The Jacobian of  $r$  is  $A$ , and  $\nabla g(r) = r$ . Under the column-gradient convention,

$$\nabla_\beta (g \circ r) = A^\top \nabla g(r) = A^\top (A\beta - y).$$

The transpose is essential. A scalar component check yields

$$\frac{\partial f}{\partial \beta_j} = \sum_{i=1}^n A_{ij} \left( \sum_{k=1}^d A_{ik} \beta_k - y_i \right),$$

which is precisely the  $j$ th entry of the matrix formula.

### 0.7.2 Why the loss is convex

Since  $h^\top A^\top Ah = \|Ah\|_2^2 \geq 0$ , the Hessian is PSD. Equivalently, the exact expansion proves the first-order inequality

$$f(\beta + h) \geq f(\beta) + \nabla f(\beta)^\top h.$$

Thus squared-error loss is convex. If  $A$  has full column rank, the Hessian is PD and  $f$  is strictly convex. When  $A$  has a nontrivial null space, moving along  $h \in \text{null}(A)$  leaves all fitted values, and hence the loss, unchanged.

**Normalization check.** For  $F(\beta) = \|A\beta - y\|_2^2$ , the gradient is  $2A^\top (A\beta - y)$  and the Hessian is  $2A^\top A$ . For  $\bar{f}(\beta) = \frac{1}{2n} \|A\beta - y\|_2^2$ , both expressions in (0.19) are divided by  $n$ . Keep this convention consistent throughout an algorithm and its analysis.

## 0.8 The normal equations and the least-squares solution

Setting the gradient to zero yields the normal equations

$$A^\top(A\hat{\beta} - y) = 0 \iff A^\top A\hat{\beta} = A^\top y. \quad (0.20)$$

If  $A \in \mathbb{R}^{n \times d}$  has full column rank  $d$ , then  $d \leq n$  and  $A^\top A \succ 0$ . Therefore,

$$\boxed{\hat{\beta} = (A^\top A)^{-1} A^\top y}. \quad (0.21)$$

This inverse formula requires full column rank. Merely having at least as many observations as coefficients does not ensure independence of the columns.

### 0.8.1 Why this stationary point is a global minimizer

Substitute a solution  $\hat{\beta}$  of (0.20) into (0.18). For every  $h \in \mathbb{R}^d$ ,

$$f(\hat{\beta} + h) = f(\hat{\beta}) + \frac{1}{2} \|Ah\|_2^2 \geq f(\hat{\beta}). \quad (0.22)$$

Thus every solution of the normal equations is a global minimizer. Under full column rank, equality requires  $h = 0$ , so the minimizer is unique. This argument establishes optimality without relying solely on the instruction “set the gradient to zero.”

### 0.8.2 A numerical line-fitting example

As a supporting example, take the observations  $(t_i, y_i) = (0, 1), (1, 2), (2, 2)$ . Then

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad y = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}, \quad A^\top A = \begin{pmatrix} 5 & 3 \\ 3 & 3 \end{pmatrix}, \quad A^\top y = \begin{pmatrix} 6 \\ 5 \end{pmatrix}.$$

Solving  $5w + 3b = 6$  and  $3w + 3b = 5$  gives

$$\hat{w} = \frac{1}{2}, \quad \hat{b} = \frac{7}{6}, \quad \hat{y} = \begin{pmatrix} 7/6 \\ 5/3 \\ 13/6 \end{pmatrix}, \quad r = y - \hat{y} = \begin{pmatrix} -1/6 \\ 1/3 \\ -1/6 \end{pmatrix}.$$

Directly,  $A^\top r = 0$  and  $\|r\|_2^2 = 1/6$ , so the minimum of the half-squared objective is  $1/12$ .

**A numerical implementation remark.** Equation (0.21) is a useful mathematical expression, but code should normally solve a linear system or use a least-squares routine instead of explicitly forming an inverse. QR and SVD methods avoid explicitly forming the normal-equation matrix. For a full-column-rank design, the ratio of largest to smallest singular values is squared when passing from  $A$  to  $A^\top A$ , which can worsen conditioning.

## 0.9 Projection geometry

### 0.9.1 Projection onto a single direction

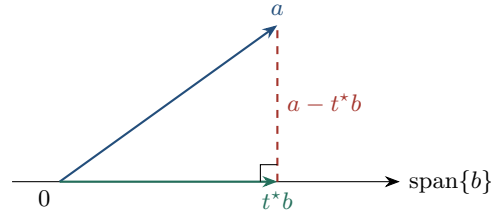
Given  $a \in \mathbb{R}^n$  and a nonzero direction  $b \in \mathbb{R}^n$ , seek the closest vector of the form  $tb$  to  $a$ . Minimizing  $\frac{1}{2} \|a - tb\|_2^2$  gives

$$\frac{d}{dt} \frac{1}{2} \|a - tb\|_2^2 = -b^\top a + t b^\top b = 0.$$

Therefore,

$$t^* = \frac{b^\top a}{b^\top b}, \quad \text{proj}_{\text{span}\{b\}} a = \frac{bb^\top}{b^\top b} a. \quad (0.23)$$

The residual  $a - t^*b$  is orthogonal to  $b$ , since  $b^\top(a - t^*b) = 0$ . The coefficient  $t^*$  can be negative: the projection is onto a line, not a ray. If  $b = 0$ , the subspace is  $\{0\}$  and the projected vector is zero, but the displayed division is not defined.



### 0.9.2 Projection onto the design column space

For a full-column-rank matrix  $A$ , the fitted response is

$$\hat{y} = A\hat{\beta} = Py, \quad P = A(A^\top A)^{-1}A^\top. \quad (0.24)$$

The normal equations imply  $A^\top(y - \hat{y}) = 0$ . Thus  $\hat{y} \in \text{col}(A)$  and  $y - \hat{y}$  is orthogonal to  $\text{col}(A)$ . For every  $z \in \text{col}(A)$ ,

$$\|y - z\|_2^2 = \|y - \hat{y}\|_2^2 + \|\hat{y} - z\|_2^2. \quad (0.25)$$

The cross term vanishes because its two vectors lie in orthogonal subspaces. This proves that  $\hat{y}$  is the closest point in  $\text{col}(A)$  to  $y$ .

The matrix  $P$  satisfies  $P^\top = P$  and  $P^2 = P$ , so it is an orthogonal projection. If  $A = U_r \Sigma_r V_r^\top$  is its compact SVD, then  $P = U_r U_r^\top$ : the fitted response retains exactly the components of  $y$  in the left-singular-vector subspace.

## 0.10 Rank deficiency and the pseudoinverse

### 0.10.1 Duplicate columns and nonunique coefficients

The final regression sketch contains duplicate feature columns. A concrete design of that form is

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ 3 & 3 & 1 \\ 4 & 4 & 1 \end{pmatrix}, \quad A\beta = (\beta_1 + \beta_2) \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \beta_3 \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}.$$

Only  $\beta_1 + \beta_2$  is identifiable. Adding  $(t, -t, 0)^\top$  to the coefficients leaves every fitted value unchanged. Here  $\text{rank}(A) = 2 < 3$ , so  $A^\top A$  is singular and the inverse in (0.21) does not exist.

For illustration, if  $y = (2, 3, 4, 5)^\top$ , every vector  $\beta = (s, 1 - s, 1)^\top$  achieves zero residual. The unique minimum-Euclidean-norm choice is  $(1/2, 1/2, 1)^\top$ , obtained by minimizing  $s^2 + (1 - s)^2 + 1$ . These responses are a supporting example, not additional measurements specified by the original sketch.

### 0.10.2 An SVD solution that works at any rank

Let  $A = U_r \Sigma_r V_r^\top$  be the compact SVD. Define the Moore–Penrose pseudoinverse by

$$A^\dagger = V_r \Sigma_r^{-1} U_r^\top \in \mathbb{R}^{d \times n}. \quad (0.26)$$

All retained singular values are positive, so the small diagonal inverse is well-defined. A least-squares solution is

$$\hat{\beta}_{\min} = A^\dagger y, \quad \hat{y} = AA^\dagger y = U_r U_r^\top y. \quad (0.27)$$

To verify optimality, split  $y$  into its orthogonal components and write

$$\|A\beta - y\|_2^2 = \|\Sigma_r V_r^\top \beta - U_r^\top y\|_2^2 + \|(I_n - U_r U_r^\top)y\|_2^2. \quad (0.28)$$

The second term is fixed, and the first is zero for  $\beta = A^\dagger y$ . Every least-squares minimizer has the form

$$\beta = A^\dagger y + z, \quad z \in \text{null}(A). \quad (0.29)$$

The two summands are orthogonal because  $A^\dagger y \in \text{col}(V_r)$ , which is orthogonal to  $\text{null}(A)$ . Hence  $\|\beta\|_2^2 = \|A^\dagger y\|_2^2 + \|z\|_2^2$ , proving the minimum-norm claim. Although coefficients may be nonunique, fitted responses remain unique. For  $A = 0$ , take  $A^\dagger = 0$  and  $\hat{y} = 0$ .

## 0.11 Taylor expansions in one variable

A derivative gives a local linear approximation; higher derivatives describe curvature and higher-order changes. If  $f$  is  $k + 1$  times continuously differentiable on an interval containing  $a$  and  $a + h$ , Taylor's theorem gives

$$f(a + h) = \sum_{j=0}^k \frac{f^{(j)}(a)}{j!} h^j + R_{k+1}(h), \quad R_{k+1}(h) = \frac{f^{(k+1)}(\xi)}{(k+1)!} h^{k+1}, \quad (0.30)$$

for some  $\xi$  between  $a$  and  $a + h$ . In particular,

$$f(a + h) = f(a) + f'(a)h + \frac{1}{2}f''(a)h^2 + \frac{1}{6}f'''(a)h^3 + R_4(h). \quad (0.31)$$

The cubic coefficient is  $1/3! = 1/6$ , not  $1/2$ . Every order has its own factorial. A finite polynomial is generally an approximation unless its remainder vanishes.

### 0.11.1 Why the remainder matters

If  $|f^{(k+1)}(x)| \leq M$  on the intervening interval, then

$$|R_{k+1}(h)| \leq \frac{M|h|^{k+1}}{(k+1)!}.$$

This turns a formal expansion into a quantitative approximation. For a quadratic polynomial, derivatives of order three and higher are zero, so its second-order Taylor expression is exact everywhere.

An infinite Taylor series equals the original function only where the remainders tend to zero. Having derivatives of all orders is not, by itself, a guarantee of equality with the Taylor series. For sine below, the derivatives are bounded and the factorial denominator supplies a direct convergence proof.

### 0.11.2 Local models for optimization

The first-order model at  $a$  is  $f(a) + f'(a)h$ , and the second-order model adds  $\frac{1}{2}f''(a)h^2$ . At a stationary point,  $f'(a) = 0$ , so the second-order term often determines the local behavior:

- $f''(a) > 0$  implies a strict local minimum.
- $f''(a) < 0$  implies a strict local maximum.
- $f''(a) = 0$  is inconclusive; higher-order terms matter.

For example,  $x^4$  and  $-x^4$  have identical first and second derivatives at zero but opposite minimum/maximum behavior there. The function  $x^3$  has neither a local minimum nor a local maximum at zero.

**Connection to least squares.** A quadratic loss is particularly simple because its curvature matrix is constant. Its second-order model is the objective itself, which is why the linear normal equations solve the problem.

## 0.12 Multivariable Taylor expansions

For  $f : \mathbb{R}^d \rightarrow \mathbb{R}$ , let  $h = y - x$ . The gradient is a vector, while the Hessian is the  $d \times d$  matrix of second partial derivatives. If  $f$  is twice continuously differentiable near  $x$ , then

$$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + o(\|h\|_2^2). \quad (0.32)$$

The middle quadratic expression is a scalar. In inner-product notation it is  $\frac{1}{2} \langle h, \nabla^2 f(x) h \rangle$ , matching the structure of the board formula.

### 0.12.1 Derivation along a line

Define  $g(t) = f(x + th)$  for  $0 \leq t \leq 1$ . The chain rule gives

$$g'(t) = \nabla f(x + th)^\top h, \quad g''(t) = h^\top \nabla^2 f(x + th) h.$$

Apply the one-dimensional Taylor formula to  $g$  at  $t = 0$  and evaluate at  $t = 1$ . This yields the multivariable formula and explains why the Hessian is multiplied by the displacement on both sides.

### 0.12.2 Third-order terms are not matrix powers

For  $f$  three times continuously differentiable near  $x$ ,

$$\begin{aligned} f(x + h) &= f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h \\ &\quad + \frac{1}{6} D^3 f(x)[h, h, h] + o(\|h\|_2^3), \\ D^3 f(x)[h, h, h] &= \sum_{i,j,k=1}^d \frac{\partial^3 f(x)}{\partial x_i \partial x_j \partial x_k} h_i h_j h_k. \end{aligned} \quad (0.33)$$

The third derivative is a trilinear form (represented by a third-order tensor). Writing  $f'''(x)h^3$  is meaningful in one dimension; in several dimensions, the contraction above specifies the intended operation.

### 0.12.3 Exact Taylor expansion of the least-squares objective

With  $f(\beta) = \frac{1}{2} \|A\beta - y\|_2^2$ ,

$$f(\beta + h) = f(\beta) + [A^\top (A\beta - y)]^\top h + \frac{1}{2} h^\top A^\top A h.$$

All higher derivatives are zero. If  $A$  has full column rank, minimizing this quadratic in  $h$  gives

$$h = -(A^\top A)^{-1} A^\top (A\beta - y), \quad \beta + h = (A^\top A)^{-1} A^\top y.$$

Thus an exact Newton step solves this unconstrained quadratic in one step, in exact arithmetic. This statement relies on an invertible Hessian and an exact linear solve; it is not a general one-step guarantee for nonlinear losses.

## 0.13 The Taylor series of sine

At zero, the derivatives of sine repeat in the cycle  $0, 1, 0, -1, 0, 1, \dots$ . Its Taylor polynomials therefore contain only odd powers:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots = \sum_{j=0}^{\infty} \frac{(-1)^j x^{2j+1}}{(2j+1)!}. \quad (0.34)$$

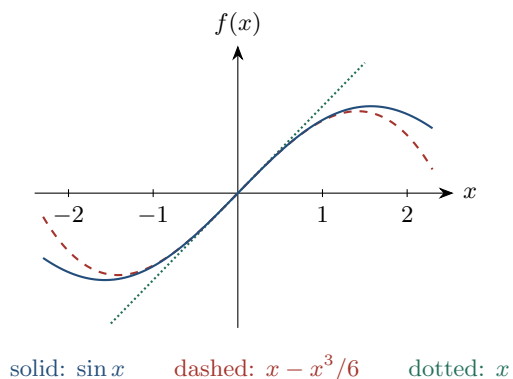
This equality holds for every real  $x$ . To justify it, every derivative of sine has absolute value at most one. Taylor's remainder after degree  $N$  is bounded by  $|x|^{N+1}/(N+1)!$ , which tends to zero for each fixed  $x$ .

### 0.13.1 Approximation accuracy

The fifth-degree polynomial is  $p_5(x) = x - x^3/6 + x^5/120$ . The sixth-degree polynomial is the same because the sixth-order coefficient is zero. Applying Taylor's theorem through degree six yields the useful bound

$$|\sin x - p_5(x)| \leq \frac{|x|^7}{7!}. \quad (0.35)$$

At  $x = 1$ ,  $p_5(1) = 101/120 \approx 0.841667$ , with error at most  $1/5040 \approx 1.9842 \times 10^{-4}$ .



All three functions agree to first order at zero. The cubic also captures the first nonzero correction to the linear approximation. The plot uses radians.

### 0.13.2 Dividing by the argument

For  $x \neq 0$ , divide (0.34) by  $x$ :

$$\frac{\sin x}{x} = 1 - \frac{x^2}{6} + \frac{x^4}{120} - \dots. \quad (0.36)$$

At  $x = 0$ , the left side is initially undefined; its continuous extension has value 1. Near zero,  $\sin x/x = 1 - x^2/6 + O(x^4)$ . The  $x^2$  coefficient will be compared with an infinite-product expansion after we establish the needed series convergence.

## 0.14 Two series with very different behavior

A sequence of terms tending to zero is necessary, but not sufficient, for an infinite series to converge. The examples in the final handwritten page make this distinction explicit.

### 0.14.1 The harmonic series diverges

Group the positive terms into blocks with endpoints at powers of two:

$$\sum_{k=1}^{2^M} \frac{1}{k} = 1 + \sum_{j=0}^{M-1} \sum_{k=2^j+1}^{2^{j+1}} \frac{1}{k}.$$

The  $j$ th block contains  $2^j$  terms, each at least  $1/2^{j+1}$ . Thus every block contributes at least  $1/2$ , and

$$\sum_{k=1}^{2^M} \frac{1}{k} \geq 1 + \frac{M}{2} \rightarrow \infty. \quad (0.37)$$

This proves divergence of  $1 + 1/2 + 1/3 + \dots$ , despite  $1/k \rightarrow 0$ . For the first few blocks, the lower bound reads

$$1 + \frac{1}{2} + \underbrace{\left(\frac{1}{3} + \frac{1}{4}\right)}_{\geq 1/2} + \underbrace{\left(\frac{1}{5} + \dots + \frac{1}{8}\right)}_{\geq 1/2} + \underbrace{\left(\frac{1}{9} + \dots + \frac{1}{16}\right)}_{\geq 1/2} + \dots$$

### 0.14.2 The reciprocal-square series converges

For every integer  $k \geq 2$ ,

$$\frac{1}{k^2} \leq \frac{1}{k(k-1)} = \frac{1}{k-1} - \frac{1}{k}.$$

The comparison telescopes, giving for  $N \geq 2$ ,

$$1 + \sum_{k=2}^N \frac{1}{k^2} \leq 1 + \sum_{k=2}^N \left( \frac{1}{k-1} - \frac{1}{k} \right) = 2 - \frac{1}{N} < 2. \quad (0.38)$$

These partial sums are increasing and bounded above, so  $S = \sum_{k=1}^{\infty} 1/k^2$  exists and  $S \leq 2$ . In fact  $S < 2$ ; retaining the strict gap at  $k = 2$  already improves the upper bound by  $1/4$ . This comparison proves convergence without using the exact value of  $S$ .

### 0.14.3 A useful tail bound

For  $N \geq 1$ , the same comparison yields

$$0 < \sum_{k=N+1}^{\infty} \frac{1}{k^2} \leq \sum_{k=N+1}^{\infty} \left( \frac{1}{k-1} - \frac{1}{k} \right) = \frac{1}{N}. \quad (0.39)$$

Thus a finite partial sum approximates the infinite sum with error at most  $1/N$ . The next section identifies the exact limit through the sine product.

## 0.15 Euler's sine product and the Basel sum

Euler's sine-product identity states that

$$\boxed{\frac{\sin x}{x} = \prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2\pi^2}\right)}, \quad (0.40)$$

with the value at zero defined continuously. We use this identity as a known theorem; its zeros alone do not constitute a proof of the product formula. An authoritative statement is NIST DLMF, Equation 4.22.1.

Pairing the factors for the positive and negative zeros explains the form appearing on the board:

$$\left(1 - \frac{x}{k\pi}\right) \left(1 + \frac{x}{k\pi}\right) = 1 - \frac{x^2}{k^2\pi^2}.$$

The product is understood as a limit of these paired partial products.

### 0.15.1 Comparing the quadratic coefficient

For a finite product, the coefficient of  $x^2$  comes from choosing the  $-x^2/(k^2\pi^2)$  term from exactly one factor and the constant term from all others. Formally, this suggests

$$\prod_{k=1}^{\infty} \left(1 - \frac{x^2}{k^2\pi^2}\right) = 1 - \frac{x^2}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} + O(x^4).$$

Comparing with (0.36) gives

$$-\frac{1}{\pi^2} \sum_{k=1}^{\infty} \frac{1}{k^2} = -\frac{1}{6}, \quad \boxed{\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}}. \quad (0.41)$$

This is the Basel sum. Because the product is infinite, the remainder must be controlled before the coefficient comparison is justified.

### 0.15.2 Why passing to the infinite product is legitimate

Put  $c_k = 1/(k^2\pi^2)$ ,  $C_N = \sum_{k=1}^N c_k$ , and  $C = \sum_{k=1}^{\infty} c_k < \infty$ , whose convergence was proved above. Expanding the finite product gives

$$P_N(x) = \prod_{k=1}^N (1 - c_k x^2) = 1 - C_N x^2 + R_N(x).$$

The sum of all products of  $j$  distinct  $c_k$  is at most  $C_N^j/j!$ , since expanding  $C_N^j$  counts each such product  $j!$  times and includes additional nonnegative terms. Consequently,

$$|R_N(x)| \leq \sum_{j=2}^{\infty} \frac{(C_N x^2)^j}{j!} \leq \frac{C^2 x^4}{2} e^{C x^2}. \quad (0.42)$$

This bound is uniform in  $N$ . Letting  $N \rightarrow \infty$  and using the sine-product identity proves  $\sin x/x = 1 - Cx^2 + O(x^4)$ . Subtract the Taylor expression, divide by  $x^2$ , and let  $x \rightarrow 0$ :  $C = 1/6$ , which establishes (0.41). This argument justifies the coefficient step without assuming the exact Basel sum in advance.

## 0.16 A compact reference and review questions

The following identities connect the lecture's topics. Dimensions and rank conditions are part of each statement.

| Object                | Identity or condition                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------|
| Matrix product        | $AB^\top = \sum_i a_i b_i^\top$ for column partitions.                                       |
| Trace                 | $\text{tr}(AB) = \text{tr}(BA)$ for compatible rectangular factors.                          |
| Squared norm          | $\ a + b\ _2^2 = \ a\ _2^2 + 2a^\top b + \ b\ _2^2$ .                                        |
| Gram matrix           | $A^\top A \succeq 0$ ; it is PD exactly at full column rank.                                 |
| Compact SVD           | $X = \sum_{i=1}^r \sigma_i u_i v_i^\top$ , $\ X\ _F^2 = \sum_{i=1}^r \sigma_i^2$ .           |
| Half-squared loss     | $\nabla f(\beta) = A^\top (A\beta - y)$ , $\nabla^2 f(\beta) = A^\top A$ .                   |
| Normal equations      | $A^\top A \hat{\beta} = A^\top y$ .                                                          |
| General-rank solution | $\hat{\beta}_{\min} = A^\dagger y$ , $\hat{y} = U_r U_r^\top y$ .                            |
| Local quadratic model | $f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top \nabla^2 f(x) h + o(\ h\ _2^2)$ . |

### 0.16.1 Review questions

1. If  $X \in \mathbb{R}^{5 \times 3}$  has rank two, give the dimensions of all factors in its full, economy-size, and compact SVDs.
2. For  $Q = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ , prove PSD and find a nonzero vector of zero quadratic form. Is  $\|\cdot\|_Q$  a norm?
3. Prove  $\langle ab^\top, cd^\top \rangle_F = (a^\top c)(b^\top d)$  for compatible vectors. Apply it to two different SVD components.
4. Derive the gradient of  $f(\beta) = \frac{1}{2} \|A\beta - y\|_2^2 + \frac{\lambda}{2} \|\beta\|_2^2$ . For  $\lambda > 0$ , explain why the Hessian is invertible even if  $A$  has duplicate columns.
5. Project  $a = (2, 3)^\top$  onto the span of  $b = (1, 1)^\top$ . Verify directly that the residual is orthogonal to  $b$ .
6. In the rank-deficient example, explain why changing the coefficients can leave the fitted responses unchanged. Which minimizer has smallest norm?
7. Bound the error of  $x - x^3/6 + x^5/120$  as an approximation to  $\sin x$  at  $x = 1/2$ , using (0.35).
8. Why is termwise coefficient comparison automatic for finite polynomials but in need of justification for an infinite product?
9. Explain why  $1/k \rightarrow 0$  does not imply convergence of  $\sum_k 1/k$ . Give a quantitative lower bound on its partial sums.